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EXAMINER	
ROY, SANJOY K	

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NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/703,514	Applicant(s) Fang et al.	
	Examiner SANJOY K ROY	Art Unit 2443	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 10/23/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,4-12,15-20 and 23-25 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,4-12,15-20 and 23-25 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 10/23/2023, 10/23/2023.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. Applicant's arguments and amendments, filed on 10/23/2023 has been entered and carefully considered. Claims 1, 12, 15 and 20 are amended. Claims 2-3, 13-14 and 21-22 are canceled. Claims 1, 4-12, 15-20, and 23-25 have been examined and rejected.

Response to Amendment and Arguments

3. Applicant's amendments and arguments filed on 10/23/2023 with respect to rejections of claims 1, 4-12, 15-20, and 23-25 have been considered but are moot in view of the new ground of rejection necessitated by applicant's amendment.

Claim Rejections - 35 USC § 103

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. Claims 1, 4-6, 8-12, 15, 20 and 23-25 are rejected under AIA 35 U.S.C. 103 as being unpatentable over Rajan et al. (U.S. PGPub 2019/0141133) in view of Kim et al. (U.S. PGPub 2018/0072250).

Rajan teaches an apparatus (Rajan, see para 0044, the in-vehicle gateway 136 handles both intra-protocol switching and inter-protocol switching CAN-to-CAN traffic as well as CAN-to-Ethernet traffic) comprising:

a first network interface circuit structured to interpret a first Ethernet data set (Rajan, see para 0051 shown in fig 11 an Ethernet frame arrives at the Ethernet switch 804 at any of the Ethernet interface port 806 with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, at 4, the CPU 814 reads the encapsulation information and performs a routing function and determines the CAN bus for the message identifier, and at 5 passes the Ethernet frame down the software protocol stack 604 converting the Ethernet frame to a CAN message protocol format, and sends the CAN message to the CAN controller for transmission)

and to transmit a second Ethernet data set (Rajan see para 0049 as shown in fig. 10, pre-configured Ethernet header encapsulation from the translation table is pre-pended to the translated CAN message to generate a Ethernet message payload sent the Ethernet switch 804 via the MAC 810 to be transmitted using the ethernet interface ports 806 and 808);

a second network interface circuit structured to transmit a first controller area network (CAN) data set (Rajan, see para 0051 shown in fig 11 at 5 when the Ethernet frame is converted to a CAN message protocol format the CAN controller transmits CAN message using the interface 138)

and to interpret a second CAN data set (Rajan, see para 0048, 0049 as shown in fig. 10, as shown in fig. a CAN message arrives at the CAN controller using the CAN interface 138 hash circuitry 826 determines an index value for the translation table, to covert a 29-bit message identifier to an 11-bit index value, the CPU uses the message identifier to perform a routing function and determine that the message destination in this example is an Ethernet port, using the 11-bit hash or received message identifier look up look up the corresponding pre-configured Ethernet header encapsulation from the translation table and pre-pended the pre-configured header to the message payload to be sent the Ethernet switch 804 via the MAC 810);

and a translation circuit structured to: determine a first message value from the first Ethernet data set and to encapsulate the first message value in the first CAN data set (Rajan, see para 0051, as shown in fig. 11 Ethernet to CAN message flow at 3, the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, the CPU 814 reads the encapsulation information and performs a routing function and determines the destination, CAN bus for the message identifier, a 5, the software protocol stack 604, which converts the Ethernet frame to a CAN message protocol format, and sends the CAN message to the CAN controller for transmission);

and determine a second message value from the second CAN data set and to encapsulate the second message value in the second Ethernet data set (Rajan, see para 0048, 0049 as shown in fig. 10, as shown in fig. a CAN message arrives at the CAN controller using the CAN interface 138 hash circuitry 826 determines an index

value for the translation table, to covert a 29-bit message identifier to an 11-bit index value, the CPU uses the message identifier to perform a routing function and determine that the message destination in this example is an Ethernet port, using the 11-bit hash or received message identifier look up look up the corresponding pre-configured Ethernet header encapsulation from the translation table and pre-pended the pre-configured header to the message payload to be sent the Ethernet switch 804 via the MAC 810);

wherein the first network interface circuit, the translation circuit, and the second network interface circuit are defined by an Ethernet switch and/or a CAN gateway both structured to be incorporated into a mobile system (Rajan, see para 0042, 0044 in-vehicle network gateway 136 handles both intra-protocol switching and inter-protocol switching, CAN-to-CAN traffic as well as CAN-to-Ethernet traffic, the in-vehicle network gateway 136 may use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN interface, the CPUs 814 may implement the control circuitry 214 with RAM 816 implementing the memory system and store the routing table 211, translation table 212, pre-defined headers 218-1-218-*n*, the software protocol stack 604, buffers 822 for storing messages, message filters 824, the hash circuitry 826, and the routing tables 828 for determining the destination interface, as an alternative implementation to routing tables stored in the RAM 816).

Rajan fails to exclusively teach

and a configuration circuit structured to modify at least one of the first network interface circuit, the translation circuit, or the second network interface circuit in

response to a configuration command value; wherein: the first network interface circuit, the translation circuit, the second network interface circuit and the configuration circuit are defined by an Ethernet switch and/or a CAN gateway both structured to be incorporated into a mobile system;

and the configuration circuit is further structured to selectively configure one or more portions of the first network interface circuit, the translation circuit, and/or the second network interface circuit to be defined by the Ethernet switch and/or the CAN gateway.

In a similar field of endeavor Kim teaches

and a configuration circuit structured to modify at least one of the first network interface circuit, the translation circuit, or the second network interface circuit in response to a configuration command value (Kim see para , see para 0092, 0093, a step S830, when the verification of the security authentication request message is completed, the controller 510 may transmit a control command to the PHY layer 520 which activate the second port P2 connected to the target communication node 600 under control of the controller 510, the target communication node 600 may receive the security authentication information from the gateway 500, controller 610 of the target communication node 600 may transmit a control command to the PHY layer 620that activate the sixth port P6 connected to the gateway 500 under control of the controller 610 so that communication between the external diagnostic apparatus 10 and the target communication node 600 may be allowed by activating the second port P2 and the sixth port P6);

wherein: the first network interface circuit, the translation circuit, the second network interface circuit and the configuration circuit are defined by an Ethernet switch and/or a CAN gateway both structured to be incorporated into a mobile system(Kim, see para 0074, 0075, controller 510 representing the configuration circuit is part of the gateway 500 where a CAN-based vehicle network 20 with plurality of CAN communication channels 530. The plurality of CAN communication channels 530 may be connected to the controller 510. The gateway 500 may exchange signals with a communication node belonging to the CAN-based vehicle network 20 via the plurality of CAN communication channels 530 and connected to a plurality of communication ports P1, P2, P3, P4, and P5 of a Phy layer switch where the first port P1 is connected to an external diagnostic apparatus 10 and the port P2 is connected to the target communication node as shown in fig. 6);

and the configuration circuit is further structured to selectively configure one or more portions of the first network interface circuit, the translation circuit, and/or the second network interface circuit to be defined by the Ethernet switch and/or the CAN gateway(Kim, see para 0075 gateway 500 may include a plurality of communication ports P1, P2, P3, P4, and P5. The first port P1 may be connected to an external diagnostic apparatus 10. The second to fifth ports P2, P3, P4 and P5 may be connected to communication nodes belonging to an Ethernet-based vehicle network 30. The gateway 500 may receive a message transmitted by the external diagnostic apparatus 10 through the first port P1. The PHY layer 520 may manage connection states of the ports P1, P2, P3, P4, and P5. The PHY layer 520 may enable or disable

connections of the ports P1, P2, P3, P4, and P5. A signal may not be transmitted through a disabled port).

It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to combine teaching of Rajan with the teaching of Kim as doing so would provide an efficient method for performing diagnostic procedures for vehicular network containing nodes using multiple different kind of network protocols communication using a common gateway from an internal and external source (Kim see para 0030).

As per claim 4

Rajan in view of Kim teaches the apparatus of claim 1, wherein the first network interface circuit is defined by the Ethernet switch incorporated into the mobile system and the second network interface circuit is defined by the CAN gateway incorporated into the mobile system (Rajan see para 0041, the gateway circuitry 802 includes an Ethernet switch 804 with ports 806 and 808. The Ethernet switch 804 connects through Media Access Control circuitry MAC 810 to a system bus 812, bus transceiver circuitry 820 includes CAN bus transceivers connected to the CAN interface 138).

As per claim 5

Rajan in view of Kim teaches the apparatus of claim 4, wherein the translation circuit is defined by the CAN gateway incorporated into the mobile system (Rajan see para 0042, the CPUs 814 implement all or part of the control circuitry 214 , while the

RAM 816 implement the memory system and store the routing table 211, translation table 212, pre-defined headers 218-1-218-*n*, the software protocol stack 604, and provide general purpose memory storage for the in-vehicle network gateway 136, the hardware acceleration circuitry 818 include, buffers 822 for storing messages, message filters 824, the hash circuitry 826, and the routing tables 828 for determining the destination interface as an alternative implementation to routing tables stored in the RAM 816).

As per claim 6

Rajan in view of Kim teaches the apparatus of claim 4, wherein the translation circuit is defined by the Ethernet switch incorporated into the mobile system (Rajan see para 0044, 0051the in-vehicle network gateway 136 use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN or other interface, the gateway circuitry 802 shapes the switch queues to match the rate of the target network or bus interface. At (3), the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame).

As per claim 9

Rajan in view of Kim teaches the apparatus of claim 1, wherein the second CAN data set comprises a service oriented communication, and wherein the translation circuit is further structured to mark the second Ethernet data set as a service oriented communication (Rajan see para 0049 the outgoing message is queued as an Ethernet

frame in the appropriate queue at the egress port or ports, the queue may be selected responsive to class of service information present in the outgoing message and specified, in the pre-configured header and detected by processing circuitry in the ingress pipeline of the Ethernet switch 804).

As per claim 10

Rajan in view of Kim teaches the apparatus of claim 9, wherein the translation circuit is further structured to apply a priority of the service oriented communication to the second Ethernet data set (Rajan see para 0032, 0044, the in-vehicle network gateway 136 may provide priority handling of messages, prioritize high priority messages over low priority message, the in-vehicle network gateway 136 may treat non-Ethernet interfaces as logical ports, and add different queues per non-Ethernet interface assigned to low and high priority traffic, CAN bus messages flowing to the Ethernet network can be aggregated by the control circuitry 214 by aggregating the message payloads with additional subsequently received message payloads into a single Ethernet packet, multiple message payloads associated with a given specific protocol, the CAN bus protocol may be destined for a common recipient, as reflected by having a common pre-defined header determined from the individual message identifiers as received, the control circuitry 214 may aggregate these message payloads into a common outgoing message by performing aggregation, until a transmission criterion or criteria are satisfied such as a message payload with greater than a pre-defined threshold level of priority is received, and this preempt low priority message aggregation).

As per claim 11

Rajan in view of Kim teaches the apparatus of claim 1, wherein the translation circuit is further structured to determine a CAN identifier of the second CAN data set, and to apply a priority to the second Ethernet data set in response to the CAN identifier(Rajan see para 0048, 0049 the CPU uses the message identifier generated from hash circuitry 826 as an index value for the translation table converting a 29-bit message identifier to an 11-bit index value to perform a routing function and determine that the message destination in this example is an Ethernet port, at the index value the 11-bit hash or received message identifier is applied to the translation table to look up the corresponding pre-configured Ethernet header to pre-pended to the message payload and the resulting outgoing message is sent to the Ethernet switch 804 via the MAC 810 with aggregation of multiple incoming message payloads placed into a single outgoing Ethernet packet that is queued as an Ethernet frame in the appropriate queue at the egress port selected responsive to class of service information present in the outgoing message and specified in the pre-configured header and detected by processing circuitry in the ingress pipeline of the Ethernet switch 804, the outgoing message the Ethernet switch 804 schedules and transmits the outgoing message out of the destination port based on the priority of the queues set by the class service information).

As per claim 12

Rajan in view of Kim teaches the method (Rajan, see para 0044, in-vehicle network gateway 136 handles both intra-protocol switching and inter-protocol switching CAN-to-CAN traffic as well as CAN-to-Ethernet traffic represented as a converged network device) comprising:

interpreting, via a first interface circuit of a converged network device (CND), a first Ethernet data set (Rajan, see para 0051 shown in fig 11 an Ethernet frame arrives at the Ethernet switch 804 at any of the Ethernet interface port 806 with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, at 4, the CPU 814 reads the encapsulation information and performs a routing function and determines the CAN bus for the message identifier, and at 5 passes the Ethernet frame down the software protocol stack 604 converting the Ethernet frame to a CAN message protocol format, and sends the CAN message to the CAN controller for transmission);

determining, via a translation circuit of the CND, a first message value from the first Ethernet data set (Rajan, see para 0051, as shown in fig. 11 Ethernet to CAN message flow at 3, the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, the CPU 814 reads the encapsulation information and performs a routing function and determines the destination, CAN bus for the message identifier);

encapsulating, via the translation circuit, the first message value in a first controller area network (CAN) data set (Rajan see para 0027, 0051, at step 5 the CPU passes the Ethernet frame down the software protocol stack 604, which converts the

Ethernet frame to a CAN message protocol format by, translation table 212 in RAM 816 may provide a lookup to map the received message to a pre-defined headers of an outgoing CAN message using pointers 216 to any number of pre-defined headers 218-1 through 218-*n*, stored in the RAM) ;

transmitting, via a second interface circuit of the CND, the first CAN data set (Rajan, see para 0051 shown in fig 11 at 5 when the Ethernet frame is converted to a CAN message protocol format the CAN controller transmits CAN message using the interface 138);

interpreting, via the second interface circuit, a second CAN data set (Rajan, see para 0047 as shown in fig. 10 CAN to Ethernet message flow, at 1 a CAN message arrives at the CAN controller through the interface 138 and is stored in the buffer 822, at 2 the message filters 824 apply to pass messages of interest to the CPU for further processing, the hash circuitry 826 determines an index value for the translation table, to covert a 29-bit message identifier to an 11-bit index value, if the length of the received message identifier is already compatible with the translation table address space, the hash circuitry 826 is skipped).

determining, via the translation circuit, a second message value from the second CAN data set (Rajan, see para 0048 as shown in fig. 10, the CPU uses the CAN message identifier to perform a routing function and determine that the message destination is an Ethernet port, using the 11-bit hash or received message identifier look up the corresponding pre-configured Ethernet header encapsulation from the translation table);

encapsulating, via the translation circuit, the second message value in a second Ethernet data set (Rajan see para 0049 at 6 the retrieved pre-configured header is pre-pended to the message payload and the resulting outgoing message is sent to the Ethernet switch 804 via the MAC 810, if aggregation is enabled, the aggregation may result in multiple incoming message payloads placed into a single outgoing Ethernet packet);

and transmitting, via the first interface circuit, the second Ethernet data set (Rajan see para 0049 at 7 the outgoing message is queued as an Ethernet frame in the appropriate queue at the egress port or ports, at 8 the outgoing message the Ethernet switch 804 schedules and transmits the outgoing message out of the destination port);

wherein the CND is disposed onboard a vehicle and defined at least in part by an Ethernet switch and/or a CAN gateway (Rajan see para 0044, in-vehicle network gateway 136 handles both intra-protocol switching and inter-protocol switching, CAN-to-CAN traffic as well as CAN-to-Ethernet traffic the in-vehicle gateway 136 may use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN or other interface).

Rajan fails to exclusively teach

the method further comprising: interpreting a configuration command value; and modifying the CND in response to the configuration command value via selectively configuring which of one or more portions of the first interface circuit, the translation circuit and/or the second interface circuit are defined at least in part by the Ethernet switch and/or the CAN gateway.

In a similar field of endeavor Kim teaches

the method further comprising: interpreting a configuration command value(Kim see para 0087-0092, as shown in fig. 8 step S810, the external diagnostic apparatus 10 transmit a diagnostic request message through the first port P1 through PHY layer 520 that require security verification, to the controller 510 Which determine an Ethernet MAC address or an IP address of the target communication node to be diagnosed from the diagnosis request message, the diagnostic message include name of the target communication node 600, the controller 510 determine the Ethernet MAC address or the IP address of the target communication node 600 using previously stored information from the name or function of the target communication node 600 included in the diagnostic request message,

a step S824, when the verification of the security authentication request message is completed, the PHY layer 520 may transmit security authentication information for the external diagnostic apparatus 10 to the target communication node 600, the controller 510 may determine which port is connected to the target communication node 600 based on the Ethernet MAC address or the IP address of the target node 600 determined in the step S812 and inform the PHY layer 520 of information on the port connected to the target communication node 600, PHY layer 520 may transmit the security authentication information to the target node 600 via the second port P2 step S830, when the verification of the security authentication message is completed, the controller 510 may transmit a control command to the PHY layer 520 to activate the second port P2 connected to the target communication node 600 under control of the controller 510)

and modifying the CND in response to the configuration command value via selectively configuring which of one or more portions of the first interface circuit (Kim see para , see para 0092, 0093, a step S830, when the verification of the security authentication request message is completed, the controller 510 may transmit a control command to the PHY layer 520 which activate the second port P2 connected to the target communication node 600 under control of the controller 510, the target communication node 600 may receive the security authentication information from the gateway 500, controller 610 of the target communication node 600 may transmit a control command to the PHY layer 620that activate the sixth port P6 connected to the gateway 500 under control of the controller 610 so that communication between the external diagnostic apparatus 10 and the target communication node 600 may be allowed by activating the second port P2 and the sixth port P6);

the translation circuit and/or the second interface circuit are defined at least in part by the Ethernet switch and/or the CAN gateway (Kim, see para 0074, 0075, controller 510 representing the configuration circuit is part of the gateway 500 where a CAN-based vehicle network 20 with plurality of CAN communication channels 530. The plurality of CAN communication channels 530 may be connected to the controller 510. The gateway 500 may exchange signals with a communication node belonging to the CAN-based vehicle network 20 via the plurality of CAN communication channels 530 and connected to a plurality of communication ports P1, P2, P3, P4, and P5 of a Phy layer switch where the first port P1 is connected to an external diagnostic apparatus 10 and the port P2 is connected to the target communication node as shown in fig. 6);

It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to combine teaching of Rajan with the teaching of Kim as doing so would provide an efficient method for performing diagnostic procedures for vehicular network containing nodes using multiple different kind of network protocols communication using a common gateway from an internal and external source (Kim see para 0030).

As per claim 15

Rajan in view of Kim teaches the method of claim 12 further comprising: generating the configuration command value external to the vehicle; and transmitting the configuration command value to the CND(Kim, see para 0084, 0098, 0099, an external diagnostic apparatus 640 may be connected to the CAN-based vehicle network by a connector with the gateway 600 as the CND, and perform diagnostic procedures for the plurality of end nodes 511 to 516 belonging to the CAN-based vehicle network, external diagnostic apparatus is connected to the gateway the external diagnostic apparatus generate a diagnostic initiation message indicating that a diagnostic procedure by the external diagnostic apparatus is to be performed at S702, diagnostic initiation message as configuration command may include at least one information element as value an an identifier of the target end node to be diagnosed by the external diagnostic apparatus and Diagnostic type value indicating the type of the diagnostic procedure such as software update procedure to be performed by the external diagnostic apparatus, diagnostic time indicating performance time of the diagnostic

procedure to be performed by the external diagnostic apparatus and sent the diagnostic message to the gateway 600).

It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to combine teaching of Rajan with the teaching of Kim and the motivation to do so would be the same as the motivation cited in relation to claim 12.

As per claim 16

Rajan in view of Kim teaches the method of claim 12 further comprising: generating a test command value external to the vehicle; transmitting the test command value to the CND; and responsive to the test command value, executing a test procedure involving a device on an Ethernet based network and/or on a CAN based network (Kim, see para 0097, 0098, as shown in fig.6 and 7, the external diagnostic apparatus connected to the gateway 600 as CND generate a diagnostic initiation message indicating that a diagnostic procedure by the external diagnostic apparatus is to be performed with an identifier of the target end node to be diagnosed by the external diagnostic apparatus, an identifier included in the diagnostic initiation message may indicate all of end nodes belonging to the vehicle network, diagnostic type indicates the type of the diagnostic procedure with diagnostic time indicates performance time);

and responsive to the test command value, executing a test procedure involving a device on an Ethernet based network and/or on a CAN based network (Kim see para 0099, 0103, gateway 600 transmit the diagnostic initiation message in a broadcast manner, internal diagnostic apparatus may receive the diagnostic initiation message

and confirm that the diagnostic procedure of the external diagnostic apparatus is to be performed based on the received diagnostic initiation message, the internal diagnostic apparatus may stop its diagnostic procedure for the target end node when it is possible to stop its diagnostic procedure for the target end node, if the diagnostic procedure for the target end node is stopped or the diagnostic procedure for the target end node is completed, the internal diagnostic apparatus may generate a diagnostic response message with an identifier of the target end node diagnosed by the internal diagnostic apparatus, diagnostic type , diagnostic result indicates a result of the diagnostic procedure performed by the internal diagnostic apparatus).

It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to combine teaching of Rajan with the teaching of Kim and the motivation to do so would be the same as the motivation cited in relation to claim 13.

As per claim 17

Rajan in view of Kim teaches the method of claim 12, wherein the first network interface circuit is defined by the Ethernet switch and the second network interface circuit is defined by the CAN gateway (Rajan see para 0041, the gateway circuitry 802 includes an Ethernet switch 804 with ports 806 and 808. The Ethernet switch 804 connects through Media Access Control circuitry MAC 810 to a system bus 812, bus transceiver 820 includes CAN bus transceivers connected to the CAN interface 138).

As per claim 18

Rajan in view of Kim teaches the method of claim 17, wherein the translation circuit is defined by the CAN gateway (Rajan see para 0042, the CPUs 814 implement all or part of the control circuitry 214 , while the RAM 816 implement the memory system and store the routing table 211, translation table 212, pre-defined headers 218-1-218-*n*, the software protocol stack 604, and provide general purpose memory storage for the in-vehicle network gateway 136, the hardware acceleration circuitry 818 include, buffers 822 for storing messages, message filters 824, the hash circuitry 826, and the routing tables 828 for determining the destination interface as an alternative implementation to routing tables stored in the RAM 816).

As per claim 19

Rajan in view of Kim teaches the method of claim 17, wherein the translation circuit is defined by the Ethernet switch (Rajan see para 0044, 0051the in-vehicle network gateway 136 use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN or other interface, the gateway circuitry 802 shapes the switch queues to match the rate of the target network or bus interface at step 3 the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame).

As per claim 20

Rajan teaches a system comprising: a vehicle comprising a first network zone comprising a Controller Area Network (CAN) based network, and a second network

zone comprising an Ethernet based network (Rajan see para 0022, 0029 in the in-vehicle network gateway 136, there are multiple different automotive network or bus interfaces 138-146, including a protocol A automotive network or a CAN bus interface connected to a CAN network representing the first network zone with an address space of possible 11-bit message identifiers for CAN messages and a protocol B automotive network or an Ethernet interface connected to a Ethernet network as the second network zone, with a list of pre-configured ethernet header for each ethernet entity with Ethernet header data, source MAC address and destination MAC address, source and destination IP address, and TCP or UDP source and destination port, there may be an instance of buffer circuitry for each different network or bus interface 138-146);

a converged network device (CND) (Rajan, see para 0044, in-vehicle network gateway 136 handles both intra-protocol switching and inter-protocol switching CAN-to-CAN traffic as well as CAN-to-Ethernet traffic represented as a converged network device) comprising:

a first network interface circuit structured to interpret a first Ethernet data set and to transmit a second Ethernet data set (Rajan, see para 0051 shown in fig 11 an Ethernet frame arrives at the Ethernet switch 804 at any of the Ethernet interface port 806 with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, at 4, the CPU 814 reads the encapsulation information and performs a routing function and determines the CAN bus for the message identifier, and at 5 passes the Ethernet frame down the software protocol stack 604 converting the Ethernet frame to a CAN message protocol format, and sends the CAN message to the CAN controller for transmission);

a second network interface circuit structured to transmit a first CAN data set and to interpret a second CAN data set (Rajan, see para 0047 as shown in fig. 10 CAN to Ethernet message flow, at 1 a CAN message arrives at the CAN controller through the interface 138 and is stored in the buffer 822, at 2 the message filters 824 apply to pass messages of interest to the CPU for further processing, the hash circuitry 826 determines an index value for the translation table, to covert a 29-bit message identifier to an 11-bit index value, if the length of the received message identifier is already compatible with the translation table address space, the hash circuitry 826 is skipped);

and a translation circuit structured to: determine a first message value from the first Ethernet data set and to encapsulate the first message value in the first CAN data set (Rajan, see para 0051, as shown in fig. 11 Ethernet to CAN message flow at 3, the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame with IEEE 1722a encapsulation that specifies a message identifier and other automotive data flow information, the CPU 814 reads the encapsulation information and performs a routing function and determines the destination, CAN bus for the message identifier);

and determine a second message value from the second CAN data set (Rajan see para 0048 as shown in fig. 10, the CPU uses the CAN message identifier to perform a routing function and determine that the message destination is an Ethernet port, using the 11-bit hash or received message identifier look up the corresponding pre-configured Ethernet header encapsulation from the translation table)

and to encapsulate the second message value in the second Ethernet data set (Rajan see para 0049 at 6 the retrieved pre-configured header is pre-pended to the message payload and the resulting outgoing message is sent to the Ethernet switch 804 via the MAC 810, if aggregation is enabled, the aggregation may result in multiple incoming message payloads placed into a single outgoing Ethernet packet);

an Ethernet switch comprising a portion of the Ethernet based network (Rajan see para 0041, the gateway circuitry 802 includes an Ethernet switch 804 with ports 806 and 808, Ethernet switch 804 connects through Media Access Control circuitry MAC 810 to a system bus 812, bus transceiver 820 includes CAN bus transceivers connected to the CAN interface 138);

a configurable edge gateway (CEG) comprising a portion of the CAN based network (Rajan see para 0035, 0036, 0041, the gateway circuitry 802 representing a configurable edge gateway includes an Ethernet switch 804 with ports 806 and 808, Ethernet switch 804 connects through Media Access Control circuitry MAC 810 to a system bus 812, system bus 812 provides a communication mechanism that interconnects the CPUs 814, the RAM 816, and the hardware acceleration circuitry 818, the MAC 810 performs bus conversion from the system bus 812 to the format of the Ethernet port interface at the Ethernet switch 804, bus transceiver circuitry 820 includes CAN bus transceivers connected to the CAN interface 138 and connected to the system bus using the hardware acceleration circuitry 818, the optional Hardware acceleration is enabled or disabled according to a global configuration parameter, or enabled or disabled on a message-by-message basis, in configuration settings stored in memory

in-vehicle network gateway 136 as the converged network device, connecting the CAN network to the Ethernet network, implements a bypass mode of hardware acceleration);

and wherein the first network interface circuit, the translation circuit, and the second network interface circuit are defined by the Ethernet switch and/or the CEG gateway (Rajan see para 0042, 0044 in-vehicle network gateway 136 handles both intra-protocol switching and inter-protocol switching, CAN-to-CAN traffic as well as CAN-to-Ethernet traffic, the in-vehicle network gateway 136 may use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN interface, the CPUs 814 may implement the control circuitry 214 with RAM 816 implementing the memory system and store the routing table 211, translation table 212, pre-defined headers 218-1-218-*n*, the software protocol stack 604, buffers 822 for storing messages, message filters 824, the hash circuitry 826, and the routing tables 828 for determining the destination interface, as an alternative implementation to routing tables stored in the RAM 816).

Rajan fails to exclusively teach and a configuration circuit defined by the Ethernet switch and/or the CEG and structured to selectively configure, in response to a configuration command value, which of one or more portions of the first network interface circuit, the translation circuit, and/or the second network interface circuit are defined by the Ethernet switch and/or the CEG.

In a similar field of endeavor Kim teaches

and a configuration circuit defined by the Ethernet switch and/or the CEG and structured to selectively configure(Kim see para , see para 0092, 0093, a step S830, when the verification of the security authentication request message is completed, the

controller 510 may transmit a control command to the PHY layer 520 which activate the second port P2 connected to the target communication node 600 under control of the controller 510, the target communication node 600 may receive the security authentication information from the gateway 500, controller 610 of the target communication node 600 may transmit a control command to the PHY layer 620that activate the sixth port P6 connected to the gateway 500 under control of the controller 610 so that communication between the external diagnostic apparatus 10 and the target communication node 600 may be allowed by activating the second port P2 and the sixth port P6),

in response to a configuration command value (Kim see para 0087-0092, as shown in fig. 8 step S810, the external diagnostic apparatus 10 transmit a diagnostic request message through the first port P1 through PHY layer 520 that require security verification, to the controller 510 Which determine an Ethernet MAC address or an IP address of the target communication node to be diagnosed from the diagnosis request message, the diagnostic message include name of the target communication node 600, the controller 510 determine the Ethernet MAC address or the IP address of the target communication node 600 using previously stored information from the name or function of the target communication node 600 included in the diagnostic request message),

the translation circuit, and/or the second network interface circuit are defined by the Ethernet switch and/or the CEG (Kim, see para 0074, 0075, controller 510 representing the configuration circuit is part of the gateway 500 where a CAN-based vehicle network 20 with plurality of CAN communication channels 530. The plurality of

CAN communication channels 530 may be connected to the controller 510. The gateway 500 may exchange signals with a communication node belonging to the CAN-based vehicle network 20 via the plurality of CAN communication channels 530 and connected to a plurality of communication ports P1, P2, P3, P4, and P5 of a Phy layer switch where the first port P1 is connected to an external diagnostic apparatus 10 and the port P2 is connected to the target communication node as shown in fig. 6).

It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to combine teaching of Rajan with the teaching of Kim as doing so would provide an efficient method for performing diagnostic procedures for vehicular network containing nodes using multiple different kind of network protocols communication using a common gateway from an internal and external source (Kim see para 0030).

As per claim 23

Rajan in view of Kim teaches the system of claim 20, wherein the first network interface circuit is defined by the Ethernet switch and the second network interface circuit is defined by the CEG (Rajan see para 0041, the gateway circuitry 802 includes an Ethernet switch 804 with ports 806 and 808. The Ethernet switch 804 connects through Media Access Control circuitry MAC 810 to a system bus 812, bus transceiver circuitry 820 includes CAN bus transceivers connected to the CAN interface 138).

As per claim 24

Rajan in view of Kim teaches the system of claim 23, wherein the translation circuit is defined by the CEG (Rajan see para 0042, the CPUs 814 implement all or part of the control circuitry 214 , while the RAM 816 implement the memory system and store the routing table 211, translation table 212, pre-defined headers 218-1-218-*n*, the software protocol stack 604, and provide general purpose memory storage for the in-vehicle network gateway 136, the hardware acceleration circuitry 818 include, buffers 822 for storing messages, message filters 824, the hash circuitry 826, and the routing tables 828 for determining the destination interface as an alternative implementation to routing tables stored in the RAM 816).

As per claim 25

Rajan in view of Kim teaches the system of claim 23, wherein the translation circuit is defined by the Ethernet switch (Rajan see para 0044, 0051 the in-vehicle network gateway 136 use the packet buffers in the Ethernet switch 804 to store packets from the Ethernet interface and then shape the traffic out to the CAN or other interface, the gateway circuitry 802 shapes the switch queues to match the rate of the target network or bus interface, in step 3 the MAC 810 receives the Ethernet frame, and the CPU 814 reads and processes the Ethernet frame).

6. Claims 7 and 8 are rejected under AIA 35 U.S.C. 103 as being unpatentable over Rajan et al. (U.S. PGPub 2019/0141133) in view of Kim et al. (U.S. PGPub 2018/0072250) in view of LIN et al. (WIPO Pub WO 2020150872).

As per claim 7

Rajan in view of Kim teaches the apparatus of claim 1, yet fails to exclusively teach wherein the translation circuit is further structured to determine a priority of a flow related to the second CAN data set, and to apply the priority of the flow to the second Ethernet data set.

In a similar filed of endeavor LIN teaches wherein the translation circuit is further structured to determine a priority of a flow related to the second CAN data set, and to apply the priority of the flow to the second Ethernet data set (LIN see page 7, lines 25-40, page 18, lines 15-29, page 19, lines 5-15mapping the value of the CAN identifier into one or more data link layer fields of the egress Ethernet frame, said fields comprising VLAN tag field, Ethernet type field, stream ID field representing a flow, source address field and destination address field, multiple ingress CAN frames can be aggregated in one egress Ethernet frame, ingress CAN frames which have similar levels of transmission priority importance can be transmitted together thus improving priority receipt in the Ethernet network, when encapsulating CAN frames in the egress Ethernet frame, each CAN ID is extracted and preserved, the priority information expressed by the CAN IDs is preserved by mapping the CAN IDs to the User priority and optionally other fields, in the switch rules, CAN frames with certain CAN IDs are switched to Ethernet interface 1, CAN frames are aggregated into Ethernet frames with a stream ID A, as shown in table 5 Aggregates CAN data field with priority 3 and stream ID A).

It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to combine teaching of Rajan in view of Kim with the teaching of LIN as doing so would provide an efficient method for protocol conversion

method for Bi-directional communication between a CAN network and a Ethernet network correctly preserving property of the prioritized transmission from source network frame in a destination network frame (LIN see page 3, lines 34-40).

As per claim 8

Rajan in view of Kim teaches the apparatus of claim 1, wherein the translation circuit is further structured to determine a priority of the second CAN data set, and to apply the priority to the second Ethernet data set LIN see page 7, lines 25-40, page 18, lines 15-29, page 19, lines 5-15mapping the value of the CAN identifier into one or more data link layer fields of the egress Ethernet frame, said fields comprising VLAN tag field, Ethernet type field, stream ID field representing a flow, source address field and destination address field, multiple ingress CAN frames can be aggregated in one egress Ethernet frame, ingress CAN frames which have similar levels of transmission priority importance can be transmitted together thus improving priority receipt in the Ethernet network, when encapsulating CAN frames in the egress Ethernet frame, each CAN ID is extracted and preserved, the priority information expressed by the CAN IDs is preserved by mapping the CAN IDs to the User priority and optionally other fields, in the switch rules, CAN frames with certain CAN IDs are switched to Ethernet interface 1, CAN frames are aggregated into Ethernet frames with a stream ID A, as shown in table 5 Aggregates CAN data field with priority 3 and stream ID A).

It would have been obvious to one of ordinary skill in the art before the effective filling date of the claimed invention to combine teaching of Rajan in view of Kim with the

teaching of LIN as doing so would provide an efficient method for protocol conversion method for Bi-directional communication between a CAN network and a Ethernet network correctly preserving property of the prioritized transmission from source network frame in a destination network frame (LIN see page 3, lines 34-40).

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, THIS ACTION IS MADE FINAL. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to SANJOY K ROY whose telephone number is (571)270-0675. The examiner can normally be reached on Mon-Fri 8:30am-5:00pm, EST. If attempts to reach the examiner by telephone are unsuccessful, the examiner's

supervisor, Nicholas R. Taylor can be reached on 571-272-3889. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272- 1000.

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